PHRON | Master Discovery Mind Map

AI Personal Finance Mentor — 0→1 Product Sprint Architecture

MASTER MAP V1.0

1. Problem Space & Origin — THEME A

- **UPI Friction:** Invisible spend in micro-txns (Zomato/Blinkit)
- **Month-End Shock:** Unconscious cash depletion
- **Spreadsheet Burnout:** Manual logs feel like a second job
- **Failure of Advice:** Generic "Faltu ka Gyan" lacks workflow

2. User Archetypes — THEME B

- **ICP v0:** Metropolitan pros (22–32 yrs, ₹35k–1.5L)
- **Aspirational Drifter:** Founder (bleeds cash, needs limits)
- **Disciplinarian:** Friend B (wants visual payoff over Excel)
- **Distressed Borrower:** Friend A (needs non-punitive state)

3. Core Engine & Recovery — THEME C

- **Deterministic Engine:** Safe-to-spend, borrow cap math
- **LLM Ingestion:** Natural language shorthand parsing
- **Dynamic Recovery:** Flex goal vs Fixed goal branching
- **Borrow Cap:** Hard max 15–20% salary advance



4. Data & Privacy Wall — THEME D

- **LLM Isolation:** Financial figures kept in Supabase DB
- **Stateless LLM:** Only passes snippet text to Gemini
- **3 Bucket Tiers:** Obligation, Discretionary, Milestone
- **DPDP Compliance:** Zero raw financial prompt exposure

5. Evolutionary Pivots — THEME E

- **Option C Persona:** Empathetic Coach vs Loss Framing
- **Optimistic Logging:** 1-tap swap chips over quizzes
- **Two-Step Setup:** 2-min onboarding to prevent bounce
- **Soft Detective:** Drift absorbed via Unallocated bucket

6. Stack & Metrics — THEME F

- **Zero-Budget Stack:** Next.js, Vercel, Supabase, Gemini
- **North Star:** WADU (Weekly Active Disciplined Users)
- **Primary Guardrail:** Unassisted accuracy score
- **3 Core Surfaces:** Dashboard, Matrix, Bucket Config

Theme 1: Problem Space, Origin & Behavioral Friction

Deconstructing why conventional personal finance tools fail urban Indian millennials

ORIGIN STORY

⚡ INVISIBLE CASHLESS FRICTION

The UPI Reality: Frictionless digital payments (UPI, GPay, Paytm) detach emotional pain from spending. Small micro-transactions on Zomato, Swiggy, and Blinkit evaporate disposable income continuously without visual friction.

📉 MONTH-END SHOCK & DRIFT

Lack of Continuous Visibility: Users experience sudden month-end insolvency. Without continuous feedback, planned savings fail and long-term milestones (e.g., festive sale phone purchase) are abandoned.

😫 SPREADSHEET EXHAUSTION

The "Second Job" Effect: Logging rows in Excel after a 10-hour workday causes high cognitive fatigue. Manual ledger keeping is usually abandoned within 1–2 weeks due to lack of immediate psychological reward.

🗣️ REJECTION OF "FALTU KA GYAN"

Generic Content Failure: Advice like "save 20%" or "stop buying coffee" fails because it provides moralizing lectures rather than operational, dynamic workflows tailored to real-world paychecks and fixed EMIs.

💬 Discussion History: Origin & Problem Framing (Q1–Q3)

> **Founder:** "I tried budgeting in Google Sheets. After three days of coming home late from work, opening a spreadsheet to enter Rs 140 for dinner felt like an uncompensated second job. I quit. By the 25th of the month, I wanted to buy a phone in the sale, but my balance was zero. Where did it go?"

> **Product Strategist:** "Existing apps either force manual entry or auto-read SMS to show charts. But showing a chart of past damage doesn't change behavior. Why build another tracking tool?"

> **Founder:** "Tracking is useless without dynamic guidance. When I slip up on a Tuesday, I don't need a red bar chart saying 'Over Budget'. I need a mentor that says: 'Okay, you spent Rs 1,200 extra today. If we trim Rs 200 from dining and Rs 150 from shopping for the next 4 days, your phone goal stays alive.'"

DECISION: Shift from Static Ledger to Closed-Loop Adaptive Mentor

Accepted Trade-off: Manual Natural Ingestion

Rejection of SMS scraping in favor of 5-second text/voice shorthand. Accepts manual initiation to guarantee privacy and cross-platform web deployment.

Operational Reality: Sprint 2 Context

Follow-up to RoamSense sprint. Built with $0 budget using OpenAI Codex for code generation, targeting urban Indian professionals with direct salaries.

Theme 2: User Personas & Behavioral Archetypes

Mapping target users from disciplined trackers to out-of-bounds distress cases

COHORT MAP

🎯 TARGET ICP V0 PROFILE

Demographics: Salaried white-collar pros in India (22–32 yrs, salary ₹35k–1.5L/mo). Operates 2–4 UPI handles, 1–2 credit cards. Non-negotiable fixed obligations (rent, loan EMIs).

🤦 ARCHETYPE 1: ASPIRATIONAL DRIFTER

The Founder Profile: Earns stable income, maintains automated SIPs (proving habit capacity), but bleeds cash on quick-commerce and food delivery. Needs visual boundaries and non-judgmental recovery.

📝 ARCHETYPE 2: DISCIPLINED TRACKER

Friend B Profile: Highly financially literate, maintains emergency buffer, uses Excel methodically. Complains spreadsheets are unrewarding and lack dynamic feedback. Testing role: Validates natural language speed.

⚠️ ARCHETYPE 3: DISTRESSED BORROWER

Friend A Profile (Out-of-Bounds): Heavy behavioral indiscipline. Holds ₹1L unstructured informal debt, overdue credit cards, chronic month-end insolvency. Exposed need for "Financial Distress Mode."

💬 Discussion History: Defining User Archetypes & Out-of-Bounds Boundary (Q3, Q21)

> **Reviewer:** "If Friend A has zero discretionary income because fixed EMIs equal his salary, standard budgeting rules break down. Safe-to-spend is negative. How does PHRON handle someone in active financial distress?"

> **Founder:** "Option A was to block him out and say 'You need debt counseling first'. But excluding the person who needs help most is an ethical failure. We chose Option B: Permissive Mode with Urgent Debt Priority."

> **System Architect:** "What changes mechanically in Permissive Distress Mode?"

> **Founder:** "The app locks all 'want' goals (no vacation or tech upgrade tanks). It tightens living allowances and automatically routes every rupee of disciplined daily surplus toward an unstructured debt repayment container."

DECISION: Implement Permissive Distress Mode (Lock Wants, Channel Surplus to Debt)

Informal Debt Accommodation

Supports non-interest, zero-schedule loans owed to friends/family without imposing artificial financial penalties or aggressive alerts.

MVP Cohort Scope

Testing strictly limited to founder and 5–10 metropolitan friends to validate core habit formation before broader public launch.

Theme 3: Core Engine, Deterministic Logic & Dynamic Recovery

Partitioning mathematical truth from linguistic intelligence

SYSTEM LOGIC

🧮 DETERMINISTIC MATH ENGINE (CODE)

Strict Application Boundary: Application code executes all ledger math: Safe-to-Spend balance calculations, bucket decrements, category burn matrices, borrow cap enforcement, and goal run-rates.

🤖 LLM LINGUISTIC LAYER (GEMINI)

Stateless Translation Utility: Parses messy Hinglish input ("450 zomato, sent 1k to Rahul") into clean JSON. Generates empathetic coaching narratives based on deterministic deltas.

🔄 FLEXIBLE VS FIXED GOAL RECOVERY

Goal-Type Branching: Overspends on *Flexible Goals* extend target dates. Overspends on *Fixed-Date Goals* (festive sales) permit salary borrowing against next month's pool.

🛑 HARD BORROW CAP GUARDRAIL

Anti-Payday Loan Loop: Future salary borrowing for fixed goals is deterministically capped at maximum 15–20% of next month's trimmable pool to prevent multi-month deficit spirals.

💬 Discussion History: Overspend Mechanics & Mathematical Division (Q5, Q16)

> **System Architect:** "Can we let the LLM calculate the recovery plan and adjust bucket allocations directly in natural language?"

> **Product Strategist:** "Absolutely not. LLMs hallucinate basic arithmetic. If an LLM calculates a budget decrement, balance integrity is destroyed. All math must be 100% deterministic code."

> **Founder:** "What happens when a user overspends by Rs 1,500 on dining? If their phone goal is in 3 weeks for Big Billion Days, pushing the date out by two weeks means missing the discount window completely."

> **System Architect:** "We introduced goal flags. Fixed-date goals allow borrowing against next month's discretionary pool, but we enforce a hard Borrow Cap in TypeScript: `MAX_BORROW = next_month_trimmable * 0.20`."

DECISION: Deterministic Math Engine + Goal-Branching + 20% Borrow Cap

LLM Prohibitions

The LLM is strictly banned from computing balances, modifying database rows directly, or offering investment/tax/legal advice.

Empathetic Recovery Response

Instead of red error alerts, the coach generates structured recovery steps: "Rough week! Let's trim Category A by ₹X and Category B by ₹Y."

Theme 4: Data Modeling, Financial Architecture & Privacy Isolation

DATA ARCHITECTURE

Structuring the database schema and safeguarding personal financial data

🧱 THREE-TIER BUCKET SCHEMA

Immutable Bucket Classes:
- *Tier 1 (Obligation):* Untouchable (Rent, EMIs, Bills).
- *Tier 2 (Discretionary):* Trimmable (Dining, Shopping).
- *Tier 3 (Milestone):* Savings Tanks (Phone, Bike).

🔐 LLM PRIVACY ISOLATION WALL

DPDP Act Compliance: Absolute balances, total income, and global debt figures are sequestered in PostgreSQL. The LLM acts as an isolated parser receiving only raw snippet strings.

📊 LEDGER & LEDGER SNAPSHOTS

Deterministic Financial Core: Tracks `monthly_net_income`, `current_liquid_balance`, `safe_to_spend_pool`, and `future_debt_obligation` across pay-cycles.

📝 TRANSACTION METADATA

Confidence & Estimates: Stores `amount`, `assigned_bucket_id`, `confidence_score` (from parser), `is_estimated` (flagging ballpark entries), and preserved `raw_text`.

💬 Discussion History: Privacy Architecture & Schema Design (Q19)

> **Product Strategist:** "If we send the user's full account snapshot in the prompt to Gemini so it has 'context', we violate India's Digital Personal Data Protection (DPDP) principles and scare privacy-conscious users."

> **System Architect:** "Option A was zero LLM (pure regex/rules). Option B was sending full financial state in system prompts. We locked Option C: Minimalist Hybrid Architecture."

> **Founder:** "How does Option C work during a spend log?"

> **System Architect:** "When you type '450 Zomato', Gemini only receives '450 Zomato' and candidate bucket names. It returns JSON `{amount: 450, bucket: 'Dining'}`. Application code receives the JSON, updates Supabase PostgreSQL, and recalculates balances locally."

DECISION: Option C Minimalist Hybrid Wall (Zero Financial Balances to LLM)

Hard Floor Constraints

Discretionary buckets enforce user-defined non-negotiable minimum floors to prevent system over-trimming essential living needs.

Google OAuth Authentication

Supabase Google OAuth selected over SMS OTP to eliminate recurring telecom gateway costs and TRAI DLT registration overhead.

Theme 5: Evolutionary Pivots, Debates & Tone Framing

The chronological evolution of product thinking across 24 discovery questions

EVOLUTIONARY ARC

🤝 PERSONA: EMPATHETIC COACH (OPTION C)

Rejection of Shame: Tested blunt loss framing (Option A) vs neutral ledger (Option B). Selected Option C: Normalizes slips, avoids moral judgment, and immediately pivots to actionable recovery.

⚡ OPTIMISTIC LOGGING PROTOCOL

Friction Reduction: Replaced interrupting modal quizzes with instant commits and 1-tap inline category-swap chips `[Change to Snacks]`. Interrogates only on zero-context input.

🚀 TWO-STEP ONBOARDING ARCHITECTURE

Setup Velocity: Step 1 (2 mins) captures Income + Fixed EMIs to output immediate Safe-to-Spend. Step 2 (Milestone expansion) triggers only after active logging across ≥3 buckets.

🕵️ SOFT DETECTIVE RECONCILIATION

Cash Drift True-Up: Friendly conversational prompts ask for ballpark recollections of missing spend. Unaccounted delta is absorbed into "Unallocated Drift" via 1-tap.

💬 Discussion History: Behavioral Psychology & Tone Selection (Q13, Q15)

> **Founder:** "When a user overspends, shouldn't we show sharp red alerts and negative numbers to 'show the damage'? Fear and shock motivate people to discipline themselves."

> **Product Strategist:** "Behavioral economics proves the opposite. Financial guilt triggers avoidance. When users see scolding interfaces after a exhausting workday, they close the tab and churn permanently."

> **Founder:** "We tested copy variants for a Rs 1,200 overspend. Option A: 'You blew your budget by Rs 1,200. Your phone goal is compromised.' Option C: 'Rough week on dining out! Don't worry, here is the exact plan to keep your phone goal on track.' Option C felt empowering rather than exhausting."

DECISION: Lock Option C (Empathetic Coach Persona) across all UI States

Setup Velocity vs Autonomy

Prevents onboarding bounce by deferring complex goal configuration until after habit baseline is established in Step 1.

Zero-Interrogation Principle

System accepts best-guess classifications rather than halting user workflow with multi-choice modal popups.

Theme 6: Stack, Metrics, Scope Boundaries & Implementation

Operational guardrails, cost constraints, and handoff rules for implementation agents

EXECUTION PLAN

💰 RS 0 / $0 CASH STACK

Free-Tier Ecosystem: Next.js (React) hosted on Vercel, Supabase (PostgreSQL + Auth), and Google Gemini API / Groq Cloud free-tier LLM inference.

📊 3-TIER METRIC HIERARCHY

- *North Star:* WADU (Weekly Active Disciplined Users).
- *Primary Trial Metric:* 2-Week Habit Persistence.
- *System Guardrail:* Unassisted Categorization Accuracy %.

📱 3 CORE UI SURFACES

Frozen Information Architecture:
1. Home Dashboard (Safe-to-Spend & Tanks).
2. Visualize Spending (Category x Week matrix).
3. Add/Edit Buckets configuration.

🚫 EXPLICIT MVP EXCLUSIONS

Strict Non-Goals: No SMS scraping/Account Aggregator APIs, no regulated stock/mutual fund advice, no third-party credit brokering, no multi-user split ledgers.

💬 Discussion History: North Star Realignment & Scope Scope Cuts (Q12, Q17–18, Q22)

> **Founder:** "Initially I thought our North Star Metric should be 'LLM Extraction Accuracy Percentage'."

> **Product Strategist:** "System accuracy is an engineering guardrail, not a North Star. An app with 99.9% categorization precision that all users abandon after 4 days has failed as a behavioral product."

> **Founder:** "Correct. The real problem is habit persistence. We redefined our North Star as Weekly Active Disciplined Users (WADU)—users who log at least once weekly while keeping their goals on track."

> **System Architect:** "We also permanently cut outbound weekly emails and peer-to-peer split ledgers to preserve our Rs 0 budget and prevent DNS/cron complexity in MVP v1."

DECISION: Adopt WADU North Star & Lock 3-Surface Navigation Architecture

Deferred to P1 Backlog

Outbound automated weekly summary emails (due to domain/cron costs) and formal P2P lending ledgers are parked post-MVP.

Codex Implementation Mandate

The PRD must preserve deterministic balance calculations, the privacy isolation wall, and 1-tap inline category swap chips.